

Transfer Learning-enabled Contactless Blood Pressure Monitoring

Zhi Wang*, Zhe Li*, Beihong Jin, Yuhui Chen, Tao Gong

Abstract—Blood pressure is one of the important vital signs for assessing cardiovascular health. Due to the operational inconvenience and poor user compliance of cuff-based blood pressure measurement solutions, contactless sensing of blood pressure, especially via microwave radars, has emerged as a research focus. However, the paramount challenge faced is the scarcity of labeled radar data, where radar data, labeled or not, can be used to derive different modalities of physiological signals. Therefore, we leverage transfer learning techniques to build a deep learning model named TransBP for contactless blood pressure monitoring. Specifically, TransBP employs convolutional neural networks for feature extraction and combines self-attention and cross-attention mechanisms to capture and integrate spatiotemporal dependencies across intra- and inter-modality into feature embeddings. TransBP is pre-trained using abundant PPG and ECG signals collected by wearable devices to learn the inherent features related to blood pressure. Then, it is fine-tuned using limited labeled radar data and a large quantity of unlabeled radar data via a three-stage strategy. By incorporating supervised training guided by the maximum mean discrepancy loss and adversarial training guided by the regressor discrepancy loss into the fine-tuning process, TransBP actively reduces the distributional difference between the source domain and the target domain, thereby improving the generalization capability. We conduct extensive experiments using three datasets, and the experimental results show that the predicted values for systolic and diastolic blood pressure are within the clinically acceptable error range, meeting the Advancement of Medical Instrumentation (AAMI) SP10 standard and achieving at least British Hypertension Society (BHS) grade B.

Index Terms—Blood pressure prediction, contactless sensing, deep learning, transfer learning.

I. INTRODUCTION

BLOOD Pressure (BP), which is defined as the pressure of circulating blood against the arterial walls, is a vital physiological indicator of the human body. BP exhibits periodic fluctuations between systolic blood pressure (SBP) and diastolic blood pressure (DBP) values within each cardiac cycle, serving as an important indicator of cardiovascular conditions [1], [2]. Notably, hypertension, as a common chronic

disease, has been identified as one of the primary causes of the development and deterioration of cardiovascular and cerebrovascular diseases [3]. According to a report published in the Lancet, the number of people aged from 30 to 79 years with hypertension worldwide has doubled from 1990 to 2019 [4]. However, the early symptoms of hypertension are often not obvious, making it difficult for individuals to perceive abnormal fluctuations of BP under daily conditions. If an individual has abnormal BP and fails to receive timely treatment, this abnormal BP may cause damage to vital organs, including the heart, brain, kidneys, and retina, potentially leading to serious consequences such as stroke, myocardial infarction, and renal failure [5]–[7]. Therefore, daily BP monitoring and timely intervention are necessary as they can significantly reduce the incidence and mortality risk of cardiovascular diseases and their complications.

Conventional methods for BP monitoring can be broadly classified into two categories: invasive methods, and noninvasive methods, including auscultation and oscillometry. Among these methods, measuring BP directly from the artery using intra-arterial catheters is the gold standard [8], [9]. Although this method provides high precision, it is highly invasive and complex to perform. The auscultatory method uses a sphygmomanometer and a stethoscope to detect Korotkoff sounds for BP estimation [10]. The oscillometric method is based on a similar principle but instead uses pressure sensors within the cuff to measure arterial pressure fluctuations [11]. While both auscultation and oscillometry are relatively simple to operate, the pressure on the limb from the cuff can induce discomfort for the individual.

To make BP measurement more convenient and comfortable, various wearable device-based solutions have been developed. The most common one is using an ambulatory blood pressure monitoring (ABPM) device that employs a cuff and a pressure sensor to record BP over 24-hour periods [12]. Although ABPM devices offer greater convenience, they provide a poor user experience. Therefore, cuffless wearable device-based solutions, such as those based on smart watches, are growing increasingly popular. Unfortunately, the fact that such a device comes into direct contact with the user's skin limits its range of applications.

In recent years, wireless sensing technology, by analyzing human-related acoustic, optical, and electrical signals, has made remarkable and substantial progress in vital sign monitoring [13]–[15]. In the context of contactless BP monitoring, microwave radars have attracted much attention due to their high distance resolution and strong anti-interference capabilities. Typically, radars are appropriately deployed to transmit

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radio frequency signals to a human body and receive signals reflected from the human body. These reflected signals are further analyzed to capture the displacement of skin or blood vessels caused by pulse wave transmission, which is used to predict BP [16]. Existing studies develop deep learning models to learn the mapping between reflected signals and BP values and predict BP from reflected signals [17]–[20]. Due to the adoption of supervised learning, existing studies demand to collect synchronized radar signals using a specific radar device for BP ground truth. We argue that this kind of data collection is not only time-consuming, labor-intensive, and costly, but also suffers from issues such as a small amount of data and potential data bias (i.e., the non-uniform distribution of the collected BP values). The latter significantly restricts the accuracy and generalization capability of BP prediction. By contrast to synchronous collection, merely collecting signals reflected from the human body is easier, and thus a greater quantity of signals can be collected. Moreover, we note that public datasets, including MIMIC [21] and VitalDB [22], have plenty of labeled signals from wearable devices. Ideally, if these data can be fully utilized, the demand for labeled radar data is expected to be significantly reduced.

In the paper, we propose to predict BP under the transfer learning framework, taking sensing data from wearable devices as the data in the source domain and received data of a ultra-wideband (UWB) radar as that in the target domain. Specifically, we can leverage public datasets with wearable sensor data that record human physiological signals, learning BP-related features from them, and then transfer this knowledge to the task of radar-based BP prediction, so as to improve the accuracy and generalization of BP prediction. Following this line of thought, we propose a transfer learning-enabled BP prediction model named TransBP that is trained using a specific training strategy. The training strategy follows the pre-training and fine-tuning paradigm, namely, we first pre-train the model using data from a public dataset, which contains photoplethysmography (PPG) and electrocardiogram (ECG) signals from wearable devices, to extract effective BP-related features. Then, we fine-tune the model in three stages to reuse and adapt the learned BP-related knowledge for BP prediction, which enables accurate BP prediction even under the constraint of limited labeled radar data.

Our contributions are summarized as follows.

(1) We build a contactless BP prediction model, where we design a pair of convolutional networks to extract features from signals of different modalities and incorporate self-attention and cross-attention mechanisms to discern spatiotemporal dependencies among features from signals of the same modality and identify the significance of features derived from auxiliary modality signals from the perspective of primary modality signals.

(2) We propose a three-stage fine-tuning strategy, where we launch supervised training with the aid of the maximum mean discrepancy loss to decrease the distribution shift between source and target domains and then perform adversarial training in the light of a regressor discrepancy loss to align feature distributions and enhance generalization in the target domain.

(3) We conduct extensive experiments on both public and self-built datasets. On the public dataset, the mean absolute error (MAE) and standard deviation (SD) for SBP are 4.75 mmHg and 6.08 mmHg, while those for DBP are 4.26 mmHg and 5.53 mmHg, respectively. On the self-built dataset, the MAE and SD for SBP are 5.02 mmHg and 6.05 mmHg, and for DBP are 4.43 mmHg and 5.61 mmHg. These results meet the Association for the Advancement of Medical Instrumentation (AAMI) SP10 standard and reach at least British Hypertension Society (BHS) grade B. Moreover, our model also outperforms six recently proposed methods, among which three fall into wearable device-based methods, and three belong to contactless device-based methods.

The rest of the paper is organized as follows. Section II introduces the related work. Section III describes the proposed system in detail, from its composition to its training strategy. Extensive experiments are conducted in Section IV to evaluate the performance of the proposed system. Finally, the paper is concluded in Section VI.

II. RELATED WORK

Our work is related to two research topics: noninvasive BP measurement and transfer learning.

A. Noninvasive BP Measurement

Noninvasive BP measurement methods can be broadly classified into three categories: cuff-based methods, cuffless wearable device-based methods, and contactless device-based methods.

1) *Cuff-based Methods*: Cuff-based BP measurement methods include auscultation and oscillometry. The auscultation method, which was introduced by the Italian physician Scipione Riva-Rocci in 1896, adopts a mercury sphygmomanometer and a stethoscope to detect Korotkoff sounds [23]. Nowadays, electronic sphygmomanometers [24] and ABPM devices [12] using the oscillometry method have been widely adopted for monitoring BP. Although cuff-based methods can accurately measure BP, prolonged and frequent measurement can induce discomfort to users.

2) *Cuffless Wearable Device-based Methods*: These methods primarily predict BP by extracting relevant features from physiological signals, such as PPG and ECG. For example, Su et al. [25] employ a Deep Recurrent Neural Network (DeepRNN) with stacked bidirectional LSTM layers and residual connections to predict BP using PPG and ECG signals, achieving RMSEs of 3.90 mmHg for SBP and 2.66 mmHg for DBP. Haddad et al. [26] extract morphological features from PPG signals and employ a multiple linear regression model to map these features to SBP and DBP. Qin et al. [27] introduce MSA-ResNet, which is based on ResNet and integrates with channel attention, to measure BP using PPG signals. MSA-ResNet achieves MAEs of 5.98 mmHg for SBP and 3.24 mmHg for DBP. Liu et al. [28] propose a parallel cross-hybrid network that integrates a convolutional neural network backbone and a Mix-Transformer backbone. Their proposed model leverages multiple physiological signals (e.g., ECG and PPG) and employs a personalized fine-tuning strategy,

achieving MAE of 3.99 mmHg and 2.12 mmHg for SBP and DBP, respectively. Tian et al. [29] extract local features and global temporal dependencies by convolutional layers and Transformer modules from PPG signals, achieving MAEs of 4.78 mmHg for SBP and 2.89 mmHg for DBP on the MIMIC-III dataset. However, wearable device-based methods require direct contact with the user's skin; therefore, they are not suitable for individuals with skin allergies, wounds, or burns. Besides, poor skin contact can lead to a significant decline in the accuracy of their measurements.

3) *Contactless Device-based Methods*: To mitigate the discomfort caused by wearable devices during monitoring, contactless BP monitoring using millimeter-wave (mmWave) and UWB radars has emerged as a growing research focus.

In early research, it was common practice to employ one or two radars to transmit signals to areas such as the chest, wrist, or neck. The reflected signals were converted into physiological signals such as pulse wave velocity (PWV) [30], pulse transit time (PTT) [31], [32], and pulse arrival time (PAT) [33], which were subsequently utilized for BP prediction. Nevertheless, in recent years, BP prediction has predominantly relied on pulse waves, which can be derived from the signals of a single radar [17]–[20], [34]–[38]. For example, Shi et al. [17] propose a mmWave radar-based BP monitoring system named mmBP, which incorporates a delay-Doppler domain feature transformation method and a temporal referential functional link adaptive filter, achieving MEs of 0.87 mmHg for SBP and 1.55 mmHg for DBP. Wang et al. [19] present an empirical mode decomposition method to decompose the phase signals reflected from the neck and propose a dual-stream fusion neural network DSFNN-BP for pulse-wave feature extraction and BP estimation. Wang et al. [20] propose a UWB radar-based contactless BP monitoring system RF-BP, which incorporates residual modules and channel attention mechanisms. Hu et al. [18] propose a contactless arterial BP waveform monitoring system, which leverages beamforming-based data augmentation and a hybrid Transformer architecture. Unfortunately, while radars are employed in BP prediction, collecting enough synchronized radar signals and BP ground truths for model training, which is time-consuming, labor-intensive, and costly, becomes a major obstacle to accurate BP prediction. What is worse, the high costs of radars hinder their practical deployment.

B. Transfer Learning

Transfer learning enables knowledge transfer from a source domain to a target domain, thereby improving the generalization performance of models in the target domain. Given the high cost of acquiring sensing data and the substantial variability across sensing devices, directly training models using limited data from the target domain often leads to poor performance. As a result, applying transfer learning to vital sign monitoring has become an important research topic.

In the field of cardiovascular monitoring, IDTL-rPPG employs an instance-based deep transfer learning, which adjusts the weight values of the blood volume pulse signals in the source domain to facilitate heart rate estimation in the target

TABLE I: Summary of key notations.

Notation	Description
$\mathbf{X}$	Extracted feature representations.
D	Dataset utilized in different training stages.
K	Kernel size.
B	Batch size.
C	Number of channels in the extracted feature representation.
S	Length of the sequence of feature representations.
P	Patch length along the temporal dimension.
N	Number of patches.
d	Dimension of the embedding.
h	Number of attention heads.
L	Number of blocks in the cross-modal attention module.
$\mathbf{O}$	Output of the attention mechanism.
$\mathbf{W}$	Learnable projection matrix.

domain, achieving motion-robust remote heart rate monitoring [39]. Fan et al. [40] propose FewShotBP, which combines a multi-modal spectro-temporal neural network (MSTNN) with a personalization adapter. In the personalization phase, MSTNN is frozen and only the adapter is fine-tuned using limited individual-specific data. Arasteh et al. [41] employ transfer learning by adding and fine-tuning five layers on top of a pre-trained VGG16, enabling respiratory and heart rate monitoring of early childhood using a UWB radar. Gautam et al. [42] propose a discriminator-based adversarial domain adaptation framework for heart rate prediction based on PPG. Transfer learning or domain adaptation is also widely used in respiration monitoring [43], [44], sleep monitoring [45]–[47] and stress prediction [48]. However, when applying transfer learning to contactless BP monitoring, no effective approach has yet been developed to tackle both distribution shift between source and target domain data and fine-grained alignment of BP values.

Compared to existing work, our work achieves accurate BP prediction with limited labeled data by effectively leveraging unlabeled data to reduce distribution shift, and enhance performance by a three-stage fine-tuning strategy, rather than introducing an extra discriminator or adding a new personalization stage for new users.

III. OUR SYSTEM

In this section, we first give an overview of our proposed BP monitoring system. Then, we offer an in-depth depiction of the modules. Following that, we delineate the training process of TransBP, which consists of two sequential phases: pre-training and three-stage fine-tuning.

A. System Overview

Our proposed system consists of two modules: a signal pre-processing module and a BP prediction module named TransBP. The overall of the proposed system is shown in Figure 1. The key notations used in this paper are summarized in Table I.

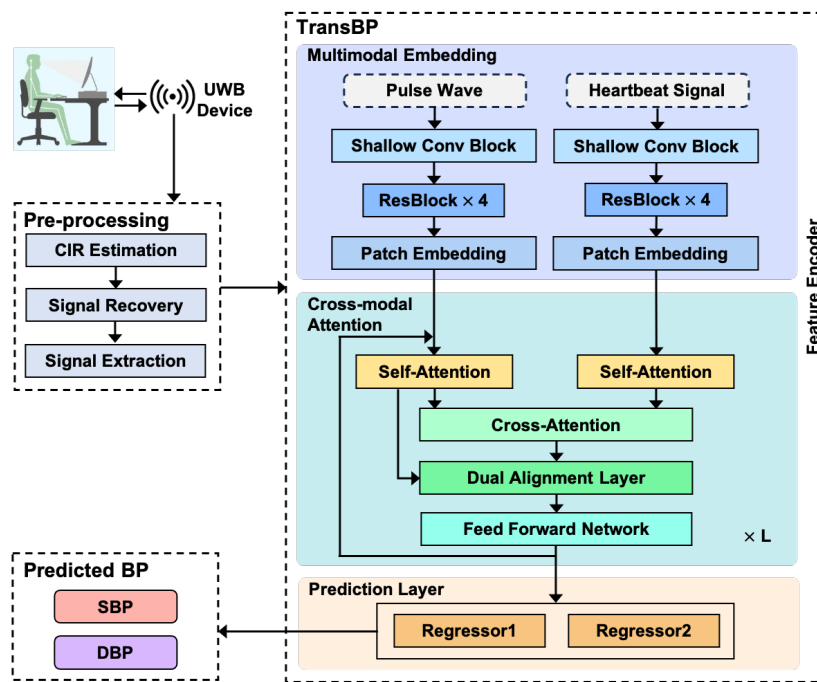


Fig. 1: Overview of the proposed system.

1) *Signal Pre-processing*: For received UWB signals, we first obtain the accurate channel impulse response (CIR) estimation. Then, we calculate the distance between the subject and the UWB device based on the amplitude of the CIR and extract phase signals from the signals reflected from the subject. Finally, we derive pulse waves and heartbeat signals from the phase signals.

2) *TransBP*: TransBP consists of a multimodal embedding module, a cross-modal attention module, and a BP prediction layer, where the first two modules are collectively referred to as the feature encoder. It is trained using labeled samples from the source domain D_s , a small set of labeled samples from the target domain D_t^l , and a larger set of unlabeled samples D_t^u . In this study, samples in the source domain refer to PPG and ECG signals from wearable devices, and samples in the target domain refer to pulse wave and heartbeat signals from a UWB radar.

- **Multimodal embedding module**. This module has two branches, each with the same convolutional neural network. These two branches extract features from the primary modality (i.e., PPG signal in the source-domain or pulse wave signal in the target-domain) and the auxiliary modality (i.e., ECG signal in the source-domain or heartbeat signal in the target-domain), and then partition the extracted features into multiple patches.
- **Cross-modal attention module**. To make the model focus on features that are highly related to BP, we design a cross-modal attention module that contains L blocks. Specifically, in each block, we first apply self-attention to the primary modality and the auxiliary modality, respectively, to capture the dependencies within the data of each modality. Then, we adopt a cross-attention mechanism to enable interaction and fusion of data from different

modalities. After that, features from the primary and auxiliary modalities are aggregated by a dual alignment layer and a feed-forward network (FFN).

- **BP prediction layer**. This layer consists of two regressors, each of which is implemented as a multilayer perceptron (MLP). The regressor accepts the features extracted by the previous modules as input and predicts SBP and DBP. In particular, while the model performs inference, only one regressor is employed to estimate BP.

B. Signal Pre-processing

The received UWB signals mainly comprise reflection of the human body and static objects within the environment. However, the presence of reflection of the static objects significantly affects the estimation of CIR in the received signals. Therefore, we apply the dynamic background subtraction method in [49] to remove the clutter caused by static reflection in the environment, retaining the signals reflected off the human body.

Then, we accumulate the CIR for each transmitted impulse signal over time and construct a CIR matrix. Based on the CIR matrix, we locate the range bin corresponding to the subject by searching for the strongest peak within a preset distance range of 1.5 m, because beyond this distance, the received signal strength attenuates substantially, making it difficult to observe the subtle pulse waves and heartbeat signals. After obtaining the distance between the subject and the device, the phase signals corresponding to that distance can be extracted.

Considering that the pulse waves are often overwhelmed by signals induced by respiration, we apply the signal recovery method in [50] to obtain accurate pulse waves. Then we apply a second-order differentiator in [51] to derive heartbeat signals.

The obtained pulse waves and heartbeat signals are then fed into TransBP to predict blood pressure.

C. Model Design

1) *Multimodal Embedding Module*: To fully capture both local features and temporal information in different physiological signals, the multimodal embedding module extracts modality-specific local features, which are subsequently mapped into vectorized representations. This mechanism not only enables comprehensive feature extraction for each modality, but also provides feature embeddings with a unified format for subsequent modules.

As shown in Figure 1, physiological signals are first fed into a shallow convolutional block, which is composed of a 1D convolutional layer (with kernel size K), a batch normalization layer, and LeakyReLU as the activation function, to extract local features. Then, four residual blocks are adopted to capture complex features. Each residual block consists of two 1D convolutional layers, batch normalization layers, and LeakyReLU activations. The skip connection is incorporated to alleviate gradient vanishing in deep networks, thus enhancing the propagation of features and improving the stability of the training.

To capture temporal dependencies in physiological signals, the extracted features are then divided into a series of non-overlapping patches. Specifically, let the shape of the feature map be $[B, C, S]$, where B is the batch size, C is the number of feature channels output by the last residual block, and S is the sequence length. A sliding window is applied with a step size of P along the temporal dimension to segment the sequence into N patches, the size of each being $[C, P]$. Each patch is then flattened, forming a sequence of patches denoted as $\mathbf{X}_{\text{patch}}$ with shape $[B, N, C \times P]$. Finally, a linear layer is used to project each patch into an embedding vector of dimension d , yielding the embedded representation $\mathbf{X}_{\text{embed}} \in \mathbb{R}^{B \times N \times d}$.

To incorporate global information, a learnable global token is introduced to the patch sequence, resulting in the input $\mathbf{X}$ for subsequent modules:

$$\mathbf{X} = [\mathbf{X}_{\text{global}}; \mathbf{X}_{\text{embed}}] \in \mathbb{R}^{B \times (N+1) \times d} \quad (1)$$

2) *Cross-modal Attention Module*: To more effectively capture cross-modal dependencies, we design a cross-modal attention module that models the dynamic interdependence between modalities, as illustrated in Figure 1. The module consists of a stack of L blocks, each of which incorporates Rotary Positional Embedding (RoPE) [52] into self-attention and cross-attention mechanisms.

While existing studies have attempted to fuse multimodal features using simple concatenation or weighted summation strategies [28], [53], [54], such methods often neglect the temporal dependencies across modalities. We argue that despite the distinct sensing mechanisms and temporal dynamics of different physiological signals, strong covariations commonly exist between them [50], [55], [56]. As RoPE encodes the positional information into the angular space of the feature vectors, allowing the model to capture relative positional

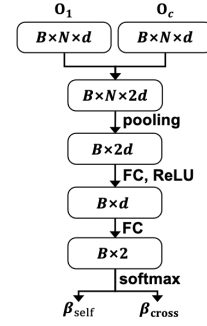


Fig. 2: Structure of the dual alignment layer.

dependencies more naturally, we adopt RoPE for positional encoding and integrate multimodal features using a cross-attention mechanism.

Specifically, let $\mathbf{X}_1$ and $\mathbf{X}_2$ be the feature sequences extracted from the primary and auxiliary modalities, respectively. They are fed into the multi-head self-attention mechanism, with RoPE encoding applied where necessary, thus yielding $\mathbf{O}_1$ and $\mathbf{O}_2$. Taking $\mathbf{O}_1$ as an example, its calculation process is as follows:

$$\mathbf{Q} = (\mathbf{X}_1 \mathbf{W}_q) \mathbf{R}_{\Theta}^d \quad (2)$$

$$\mathbf{K} = (\mathbf{X}_1 \mathbf{W}_k) \mathbf{R}_{\Theta}^d \quad (3)$$

$$\mathbf{V} = \mathbf{X}_1 \mathbf{W}_v \quad (4)$$

$$\mathbf{O}_1^i = \text{softmax}\left(\frac{\mathbf{Q}_i^T \mathbf{K}_i}{\sqrt{d/h}}\right) \mathbf{V}_i, i = 1, \dots, h \quad (5)$$

$$\mathbf{O}_1 = \text{concat}(\mathbf{O}_1^1, \mathbf{O}_1^2, \dots, \mathbf{O}_1^h) \mathbf{W}_o \quad (6)$$

where d is the feature dimension, h is the number of attention heads, $\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i$, and $\mathbf{O}_1^i$ represent the query, key, value, and output for the i -th head, $\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v$, and $\mathbf{W}_o$ are the projection matrices.

The rotation matrix corresponding to the feature at the p -th position is given as follows:

$$\mathbf{R}_{\Theta, p}^d = \begin{pmatrix} \alpha_1 & & & \\ & \alpha_2 & & \\ & & \ddots & \\ & & & \alpha_{d/2} \end{pmatrix} \quad (7)$$

$$\alpha_i = \begin{pmatrix} \cos(p\theta_i) & -\sin(p\theta_i) \\ \sin(p\theta_i) & \cos(p\theta_i) \end{pmatrix} \quad (8)$$

where the set of rotation angles is defined as $\Theta = \{\theta_i = 1000^{-2(i-1)/d} \mid i = 1, \dots, d/2\}$.

The calculation of $\mathbf{O}_2$ is similar but with distinct parameters. Subsequently, cross-attention is computed by taking $\mathbf{O}_1$ as the key and value, and $\mathbf{O}_2$ as the query, yielding $\mathbf{O}_c$.

After obtaining the outputs from the self-attention and cross-attention, we combine $\mathbf{O}_1$ and $\mathbf{O}_c$ to obtain a more comprehensive representation. Considering that the outputs of self-attention and cross-attention contribute differently to BP, we adopt an adaptive alignment layer to learn their significance, as shown in Figure 2. Specifically, we first concatenate two representations along the feature dimension and perform

global mean pooling along the sequence length to obtain a mean vector:

$$\mathbf{X}_{\text{concat}} = [\mathbf{O}_1, \mathbf{O}_c] \quad (9)$$

$$\mathbf{X}_{\text{mean}} = \frac{1}{N+1} \sum_{i=1}^{N+1} \mathbf{X}_{\text{concat}}(i, :) \quad (10)$$

Next, we apply two fully connected layers to implement dual alignment. The first layer suppresses less useful features, and the second layer outputs the weights of $\mathbf{O}_1$ and $\mathbf{O}_c$, which are denoted by β_{self} and β_{cross} , respectively. The output of dual alignment layer $\mathbf{X}_f$ can be calculated as follows:

$$\mathbf{X}_f = \beta_{\text{self}} \mathbf{O}_1 + \beta_{\text{cross}} \mathbf{O}_c \quad (11)$$

The output $\mathbf{X}_f$ is further processed by a position-wise FFN consisting of two FC layers.

Finally, the output of the cross-modal attention module is fed into the prediction layer.

3) *Model Optimization*: The model has different optimization objectives at different phases.

In the pre-training phase, the model, which is configured so that two structurally identical regressors share parameters, takes the labeled samples from D_s as input and is optimized using the loss of BP prediction over the source domain. We adopt Huber loss as the basic BP prediction loss. In addition, to balance the contributions of SBP and DBP, a task-specific weight coefficient λ is introduced. In brief, the loss for pre-training is as follows:

$$\mathcal{L}_{\text{BP}}^s = \frac{\mathbb{E}_{x,y \sim D_s} [\mathcal{L}_{\text{Huber}}(y, \hat{y}_1) + \mathcal{L}_{\text{Huber}}(y, \hat{y}_2)]}{2} \quad (12)$$

$$\mathcal{L}_{\text{Huber}}(y, \hat{y}) = \lambda \cdot \text{Huber}(y_{\text{SBP}}, \hat{y}_{\text{SBP}}) + \text{Huber}(y_{\text{DBP}}, \hat{y}_{\text{DBP}}) \quad (13)$$

where x represents the input sample, $y = \{y_{\text{SBP}}, y_{\text{DBP}}\}$ represent the ground truth, $\hat{y}_1$ and $\hat{y}_2$ represent the predicted BP values of the first and second regressors, respectively.

In the fine-tuning phase, the two regressors are set to no longer share their parameters, each of which updates its parameters independently. Furthermore, the model takes as input the labeled samples from D_s and D_t^l and the unlabeled samples from D_t^u , with the optimization objectives being loss of BP prediction $\mathcal{L}_{\text{BP}}$, loss of marginal distribution alignment $\mathcal{L}_{\text{MMD}}$, and loss of conditional distribution discrepancy $\mathcal{L}_{\text{diff}}$, respectively.

(1) The loss of BP prediction $\mathcal{L}_{\text{BP}}$ is defined as:

$$\mathcal{L}_{\text{BP}} = \mathcal{L}_{\text{BP}}^s + \mathcal{L}_{\text{BP}}^t \quad (14)$$

where $\mathcal{L}_{\text{BP}}^t$ is the loss of BP prediction over the set of labeled samples from the target domain D_t^l .

(2) Maximum Mean Discrepancy (MMD) [57] is a non-parametric statistical measure based on the distance in a Reproducing Kernel Hilbert Space (RKHS). Its core idea is to compare the mean embeddings of two distributions in a high-dimensional kernel space to quantify the distributional discrepancy between the source and target domains. When the distributions are identical, their mean embeddings converge, and the MMD value approaches zero.

Given the datasets from the source domain and target domain, the loss of marginal distribution alignment $\mathcal{L}_{\text{MMD}}$ is computed as:

$$\mathcal{L}_{\text{MMD}} = \|\mathbb{E}_{x \sim D_s} [\phi(x)] - \mathbb{E}_{x' \sim D_t} [\phi(x')]\|_{\mathcal{H}}^2 \quad (15)$$

where $D_t = D_t^l \cup D_t^u$, $\phi(\cdot)$ denotes the mapping function that projects the input data into an RKHS $\mathcal{H}$.

In practice, a kernel function $k(x, x') = \langle \phi(x), \phi(x') \rangle_{\mathcal{H}}$ is used to implicitly perform this mapping. In this work, we adopt a multi-scale Gaussian kernel, which is defined as:

$$k(x, x') = \sum_{i=1}^{n_{\sigma}} \exp\left(-\frac{\|x - x'\|^2}{\sigma_i}\right) \quad (16)$$

where the number of kernels n_{σ} is set to 5 and the bandwidths σ_i are determined adaptively from the data. Specifically, the base bandwidth σ_{base} is computed, following the approach described in [58], using the median pairwise squared distance among all samples. The final kernel set is defined as $\{\sigma_i\}_{i=1}^{n_{\sigma}} = \{\frac{1}{4}\sigma_{\text{base}}, \frac{1}{2}\sigma_{\text{base}}, \sigma_{\text{base}}, 2\sigma_{\text{base}}, 4\sigma_{\text{base}}\}$, covering a range of fine-grained to coarse-grained distances. As a result, the used MMD loss is as follows:

$$\mathcal{L}_{\text{MMD}} = \mathbb{E}_{x, x' \sim D_s} [k(x, x')] - 2\mathbb{E}_{x \sim D_s, x' \sim D_t} [k(x, x')] + \mathbb{E}_{x, x' \sim D_t} [k(x, x')] \quad (17)$$

(3) Conditional distribution discrepancy loss $\mathcal{L}_{\text{diff}}$ is defined as the output discrepancy between the two regressors on the unlabeled target domain samples. It can be maximized to encourage divergent predictions and help uncover regions with modeling ambiguities that reflect conditional distribution mismatch under the same inputs. Moreover, minimizing this discrepancy can guide the feature encoder to reduce such mismatch and achieve conditional distribution alignment.

The discrepancy $\mathcal{L}_{\text{diff}}$ is measured using the ℓ_1 distance between the outputs of the two regressors:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{x, y \sim D_t^u} [\|\hat{y}_1 - \hat{y}_2\|] \quad (18)$$

D. Model Training

Figure 3 outlines the training process of TransBP.

1) *Pre-training Phase*: In the pre-training phase, the model is only trained over the labeled source-domain dataset D_s . The objective is to minimize the supervised loss $\mathcal{L}_{\text{BP}}^s$ on the source domain. Through this phase, the model is expected to learn BP-related features and accurately predict BP, given a source-domain sample as input.

2) *Fine-tuning Phase*: In the fine-tuning phase, the model is adapted to the target domain using a small amount of labeled samples and a great quantity of unlabeled samples via a three-stage strategy, incorporating the supervised loss, loss of marginal distribution alignment, and loss of conditional distribution discrepancy, as illustrated in Figure 4.

Stage 1. The feature encoder and two regressors are jointly trained to ensure accurate BP predictions for labeled samples from the source domain and target domain. Simultaneously, marginal distribution alignment is achieved using the MMD

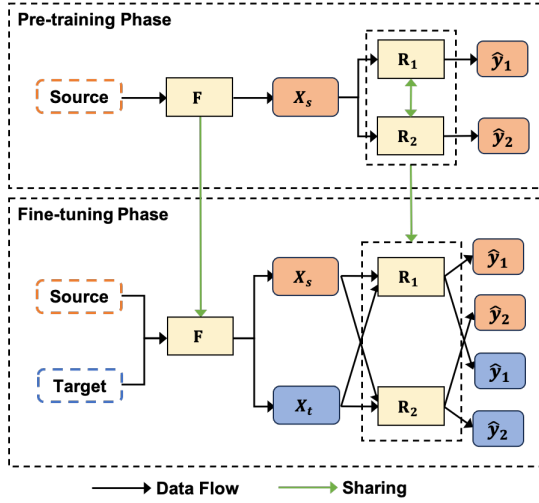


Fig. 3: Illustration of model training process. F denotes the feature encoder, which maps the input signals into domain-specific features (X_s for the source domain and X_t for the target domain). R_1 and R_2 denote the regressors.

between features derived from the source-domain samples and those derived from target-domain samples. The objective for this stage is $\min(\mathcal{L}_{BP} + \gamma \cdot \mathcal{L}_{MMD})$, where the weight coefficient γ is defined as $\gamma = \frac{10}{1+e^{-r/\tau}}$, with r denoting the current iteration and τ being a scaling constant. γ gradually increases from 0 to 10 over training iterations. This stage is crucial for establishing a stable, meaningful latent representation and learning consistent mappings for BP prediction. As illustrated on the right side of Figure 4, this stage reduces marginal distribution discrepancies between the source and target domains.

IV. EXPERIMENTAL EVALUATION

In this section, we conduct extensive experiments to answer the following Research Questions (RQs):

- **RQ1:** How well does TransBP perform in terms of BP prediction accuracy? Can it be applied to continuous BP monitoring? Does it surpass existing BP prediction models?
- **RQ2:** What contributions do different modules in TransBP make to the performance?
- **RQ3:** How do the number of training samples in the target domain and different stages in the fine-tuning phase affect performance?

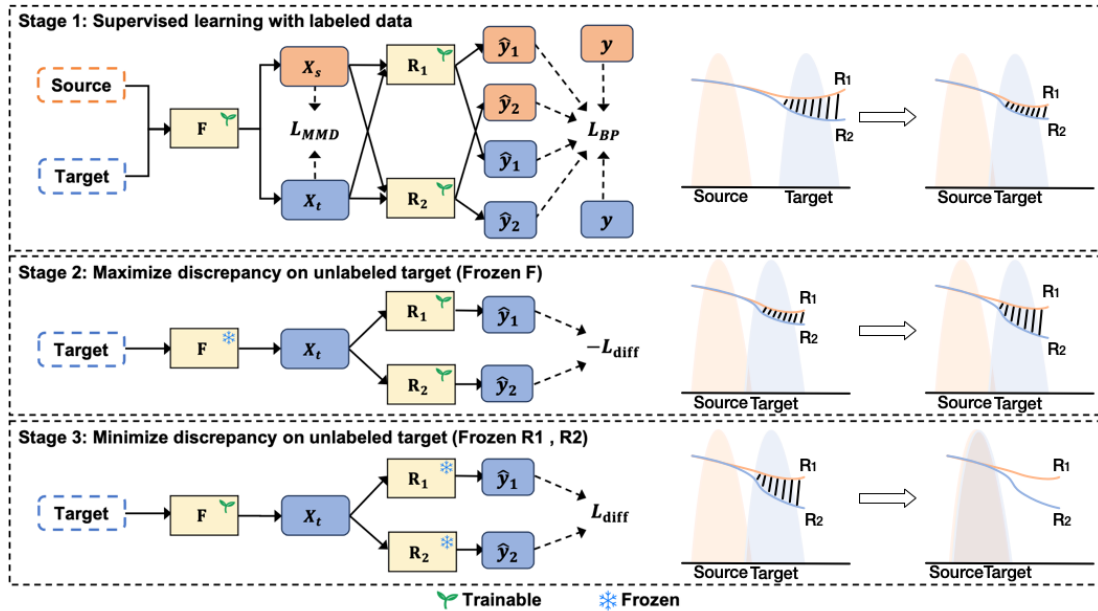


Fig. 4: Illustration of the three-stage strategy during the fine-tuning phase. Two half-ellipses represent the feature spaces of the source and target domains, respectively. The curves denote the performance of the regressors. Overlapping curves correspond to identical values for the two regressors, whereas separated curves indicate distinct values.

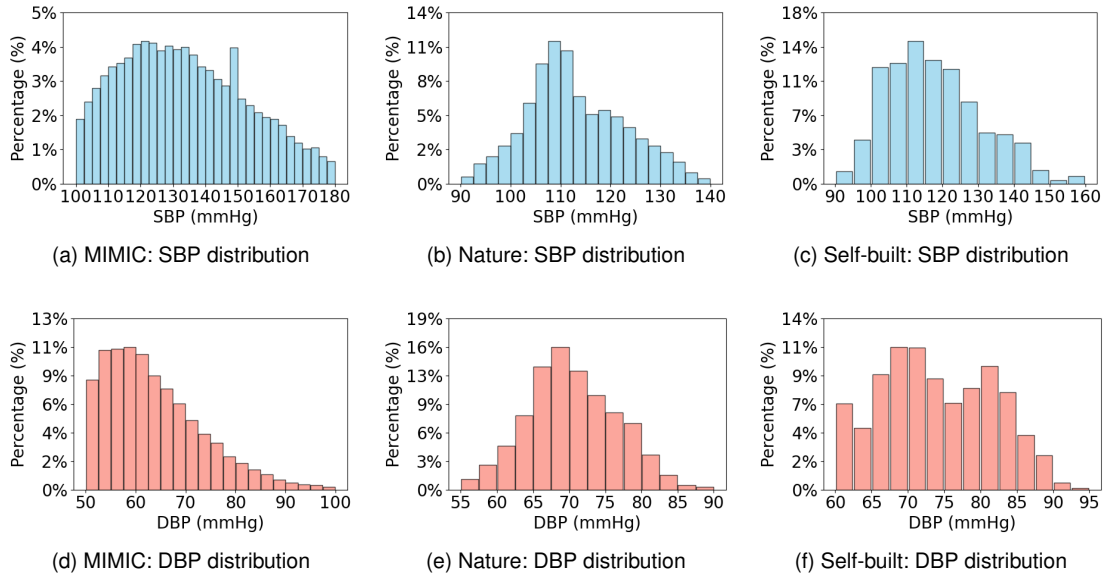


Fig. 5: SBP and DBP distributions across three datasets.

- **RQ4:** What are the effects of the important hyperparameters in TransBP on performance?
- **RQ5:** To what extent does the system exhibit robustness? Is it compatible with subjects of different ages or body mass indexes (BMIs), as well as scenarios with varying subject-device distances?

A. Experimental Setup

1) *Datasets:* We choose three datasets: two publicly available datasets (MIMIC-II dataset and the Nature dataset) and one self-built dataset.

MIMIC-II dataset [21]: It is developed by the Massachusetts Institute of Technology. In this paper, we adopt a subset of the MIMIC-II dataset [59], which contains physiological signals from approximately 12,000 subjects. This data subset includes multi-channel physiological signal recordings such as ECG, PPG, and invasive arterial blood pressure (ABP), sampled at 125 Hz. The distribution of BP corresponding to this data subset is shown in Figures 5a and 5d.

Nature dataset [60]: It is released by the research team from Friedrich-Alexander University Erlangen-Nürnberg and contains synchronized recordings of radar signals and continuous BP values from 30 healthy volunteers. All signals are recorded at a sampling frequency of 2000 Hz. The distribution of BP in Nature dataset is shown in Figures 5b and 5e.

Self-built dataset: We collect synchronized UWB signals and BP ground truth from 70 subjects (40 female, 30 male), aged from 25 to 80, including 13 diagnosed with hypertension. Before the experiments, each participant is informed about the measurements. We have obtained written consent from all participants, and our Institutional Review Boards have approved the entire research activities. For data collection, a UWB device, i.e., XeThru X4M05 with a center frequency of 7.3 GHz and a bandwidth of 1.4 GHz, is placed on the desk, and the subject sits 0.3–0.6 m away from the device.

TABLE II: British Hypertension Society grading criteria.

Grade	≤ 5 mmHg	≤ 10 mmHg	≤ 15 mmHg
Grade A	60%	85%	95%
Grade B	50%	75%	85%
Grade C	40%	65%	75%
Grade D	<40%	<65%	<75%

The signals reflected from the human body are recorded at a frequency of 200 Hz. Simultaneously, the OMRON HEM-7132 is employed to obtain the ground truth of BP. Each data collection lasts approximately 40 seconds. Ultimately, we collect 60-hour signals and corresponding BP ground-truth values. The distribution of BP corresponding to our self-built dataset is shown in Figures 5c and 5f.

All signals from these three datasets are segmented into 5-second clips. Additionally, signals from the Nature dataset and the self-built dataset are downsampled to 125 Hz. After that, each of these two datasets is split into training, validation, and test sets by a ratio of nearly 7:2:1, with the constraint that each individual's samples are entirely assigned to one subset to avoid data leakage across partitions.

2) *Evaluation Metrics:* We employ mean error (ME), MAE, and SD to evaluate the accuracy of BP prediction. To ensure the comparability and dependability of the performance results, we also adopt international standards, including AAMI SP10 standard [61] and BHS grading criteria [62]. The former specifies that the MEs of predicted SBP and DBP should be within ± 5 mmHg, and the SD should be less than 8 mmHg. The latter requires separately calculating the percentage of BP measurement errors that fall within ± 5 mmHg, ± 10 mmHg, and ± 15 mmHg, then classifying the performance into grades A through D based on predefined thresholds, as shown in Table II.

3) *Implementation Details:* We implement TransBP in PyTorch. In the multimodal embedding module, the convolution

kernel size K is set to 13, the patch length P is set to 25, the number of feature channels C is set to 16, the sequence length S is set to 625, and the embedding dimension d is set to 128. In the cross-modal attention module, L , i.e., the number of blocks, and h , i.e., the number of attention heads in a block, are set to 4 and 8, respectively. The task-specific loss weight coefficient $\lambda_{\text{SBP/DBP}}$ is set to 9. TransBP contains about 0.93M trainable parameters (occupying approximately 3.74 MB in FP32). On NVIDIA GeForce RTX 4070 Ti SUPER, the average inference latency is 0.2 s for a 5-second signal.

In the pre-training phase, the model is trained on the MIMIC-II dataset. In the fine-tuning phase, the MIMIC-II dataset serves as the data from the source domain, while the Nature dataset and the self-built dataset are treated as the data from the target domain, respectively. Most importantly, during the fine-tuning, only 20% of the samples from target domains are employed as labeled data with the remaining 80% serving as unlabeled data. To be specific, the number of labeled samples is approximately 2300 (covering 4 subjects) for the Nature dataset, or 5000 (covering 12 subjects) for the self-built dataset.

During the pre-training phase, the Adam optimizer is adopted with an initial learning rate of $1e-3$. A StepLR learning rate scheduler is used to reduce the learning rate by a factor of 0.8 every 5 epochs. The batch size is set to 128, and training is conducted for 120 epochs. In the fine-tuning phase, the Adam optimizer is also used, with an initial learning rate of $1e-4$. The batch size remains 128, and the number of training epochs is set to 60.

B. Performance Comparison (RQ1)

1) *Overall Performance*: To ensure a robust evaluation, we conduct ten-fold cross-validation independently on both the Nature dataset and our self-built dataset. The BP prediction performance is thoroughly evaluated using multiple metrics, including ME, MAE, SD, the AAMI SP10 standard, and the BHS criteria.

Detailed results of TransBP are presented in Table III. On Nature and self-built datasets, predictions made by TransBP for SBP and DBP both meet the requirements of the AAMI SP10 standard. In accordance with the BHS criteria, SBP predictions over the self-built dataset are classified as Grade B, while other predictions demonstrate superior precision, achieving Grade A.

TABLE III: Prediction performance and BHS grades of TransBP.

TransBP (mmHg)	ME	MAE	SD	≤ 5	≤ 10	≤ 15	BHS Grade
Nature Dataset							
SBP	0.07	4.75	6.08	60.9%	90.2%	98.2%	A
DBP	0.05	4.26	5.53	66.7%	92.9%	98.5%	A
Self-built Dataset							
SBP	0.92	5.02	6.05	54.6%	88.2%	97.4%	B
DBP	0.50	4.43	5.61	63.2%	92.7%	99.8%	A

TABLE IV: Performance comparison across BP ranges.

TransBP BP Range	MAE (mmHg)		SD (mmHg)		BHS Grade	
	SBP	DBP	SBP	DBP	SBP	DBP
Nature Dataset						
<110 mmHg	5.03	4.67	4.48	5.31	B	A
110-120 mmHg	2.22	3.27	2.85	4.06	A	A
120-130 mmHg	3.28	4.10	2.92	4.41	A	A
>130 mmHg	5.50	4.18	4.66	4.73	B	A
Self-built Dataset						
<110 mmHg	5.97	5.10	4.92	5.61	B	B
110-120 mmHg	3.94	4.53	4.90	5.83	A	A
120-130 mmHg	4.43	5.36	4.99	6.08	A	B
>130 mmHg	5.81	3.94	5.53	5.15	B	A

In addition, we draw the Bland-Altman plots of SBP and DBP predicted by TransBP on the Nature and self-built datasets, as shown in Figure 6 and Figure 7. The black solid line indicates the ME, and the red dashed lines represent the limits of agreement. Over 95% of the data points lie within the agreement limits, indicating strong consistency between the predicted BP values and the ground truth.

2) *Performance across Different BP Ranges*: To investigate the generalization capability of TransBP, we analyze its performance across different SBP ranges on the Nature and self-built datasets. As shown in Table IV, TransBP maintains stable prediction accuracy across a wide range of BP levels.

On the Nature dataset, TransBP achieves the lowest error in the 110–120 mmHg range. The MAE slightly increases in the extreme ranges, particularly for SBP < 110 mmHg and SBP > 130 mmHg, which is as expected due to the reduced sample size in these ranges. The range of variation in standard deviations is relatively small, indicating consistent prediction without notable overfitting to any single BP range.

On the self-built dataset, a similar trend is observed.

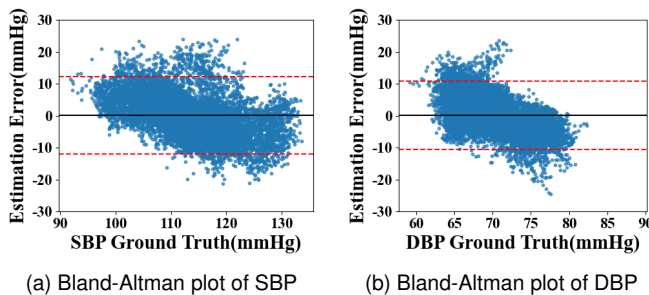


Fig. 6: Bland-Altman plots of BP on Nature dataset.

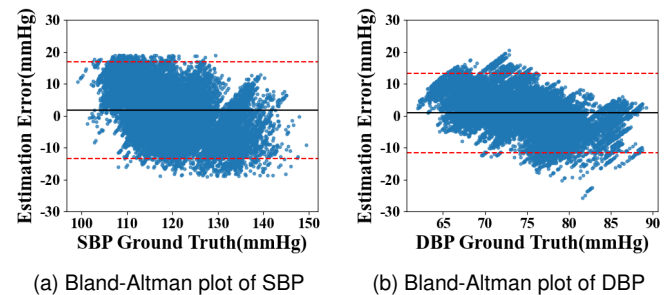


Fig. 7: Bland-Altman plots of BP on Self-built dataset.

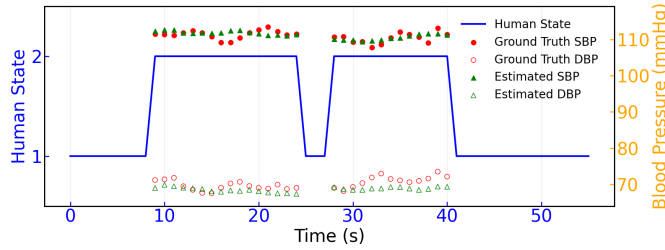


Fig. 8: Continuous monitoring of BP.

TransBP achieves its best performance in the 110–120 mmHg range for SBP and in the >130 mmHg range for DBP. These prediction errors are comparable to those on the Nature dataset, demonstrating strong cross-dataset generalization.

Overall, these results confirm that TransBP exhibits effective generalization across different BP ranges and datasets, with any performance decline in extreme ranges largely attributable to data imbalance. This reveals the potential of TransBP for practical BP monitoring across diverse populations.

3) *Continuous Monitoring of BP*: To evaluate the performance of TransBP in continuous monitoring scenarios, we conduct an experiment using a 60-second signal as input of the trained TransBP. Figure 8 shows the subject's states and the predicted BP. State 1 denotes periods when the subject is either outside the sensing range or undergoing body motion, whereas State 2 denotes stationary periods that are suitable for BP estimation. It can be seen that TransBP provides accurate and stable BP estimates in a continuous monitoring scenario.

4) *Comparison with Existing Models*: To comprehensively evaluate the performance of TransBP, we compare its performance with several existing BP prediction models. We choose as comparison models three wearable device-based models, namely DeepRNN-4L [25], MSA-ResNet [27], and PCTN [29], and three contactless device-based models, namely DSFNN-BP [19], RF-BP [20], and mmRBP [63]. Since these models are not open-sourced, we replicate the code of these models according to their published papers.

For a fair comparison with wearable device-based models, we first train these models using data from the source domain, and then train using 20% of the labeled samples of the target domain in the training set, where the signals from the Nature dataset and the self-built dataset are preprocessed to extract pulse waveforms before they are sent into these models. We conduct ten-fold cross-validation on all comparison models. Results are listed in Table V.

As shown in Table V, TransBP outperforms three wearable device-based models. This highlights the necessity of designing new models rather than directly adapting wearable device-based methods for contactless BP monitoring. Meanwhile, TransBP outperforms DSFNN-BP and RF-BP, two contactless device-based models that do not employ transfer learning. In addition, under the same training conditions, TransBP outperforms mmRBP, the model using transfer learning. This is presumably because mmRBP employs a relatively simple transfer learning strategy, which can exhibit good performance in the case of a small dataset (i.e., 15 subjects) but ultimately hinders its generalization. Furthermore, we conduct a

TABLE V: Performance comparison.

Method	MAE (mmHg)		SD (mmHg)		BHS Grade	
	SBP	DBP	SBP	DBP	SBP	DBP
Nature Dataset						
TransBP	4.75	4.26	6.08	5.53	A	A
DeepRNN-4L [25]	7.86	5.17	10.12	6.65	C	B
MSA-ResNet [27]	7.96	6.17	10.24	7.93	D	B
PCTN [29]	7.05	5.08	9.15	5.89	C	B
DSFNN-BP [19]	9.04	5.25	11.67	6.74	D	B
RF-BP [20]	7.78	5.89	10.17	6.94	C	C
mmRBP [63]	7.03	5.22	8.49	6.52	C	B
Self-built Dataset						
TransBP	5.02	4.43	6.05	5.61	B	A
DeepRNN-4L [25]	6.99	5.93	8.51	7.01	D	C
MSA-ResNet [27]	7.59	5.76	9.16	7.08	D	C
PCTN [29]	7.15	5.98	8.64	6.95	C	C
DSFNN-BP [19]	7.56	5.95	8.73	7.10	D	C
RF-BP [20]	6.96	5.49	8.33	6.69	C	C
mmRBP [63]	6.86	5.56	8.39	6.71	C	B

Wilcoxon signed-rank test on the performance of our model and each comparison method, while setting a significance level to 0.05. All p -values are less than 0.05, which indicates that there are statistically significant improvements over the comparison methods.

C. Ablation Study (RQ2)

In TransBP, we propose two key modules to enhance the accuracy of BP prediction: (1) a multimodal embedding module with a dual-branch structure that models multiple modalities separately, (2) a cross-modal attention module to improve the modeling of complementary information between multimodal signals.

To observe the effectiveness of these modules, we conduct an ablation study. We construct two variants of TransBP. The variant T1 replaces the dual-branch structure in the multimodal embedding module with a single-branch structure, where primary modality and auxiliary modality are concatenated as the initial input, resulting in a single fused feature map. Through dedicated convolutional layers, the fused feature map is then separated into distinct feature maps corresponding to the primary and auxiliary modalities, which are subsequently fed into downstream modules. The variant T2 removes the cross-attention mechanism in the cross-modal attention module. Thus, the feature embeddings from the self-attention mechanisms of the primary and auxiliary modalities are concatenated directly through the dual alignment layer and processed through fully connected layers to predict BP.

The experimental results are shown in Table VI. Experimental results of variant T1 show a more pronounced performance decrease, where on the Nature dataset, the MAE of SBP increases by 40.4% and SD by 38.9%. These results show that the dual-branch design plays a critical role in preserving modality-specific characteristics. By processing primary modality and auxiliary modality separately, TransBP retains unique physiological patterns of signals that would be diluted in a single-branch setup. On the other hand, experimental results of variant T2 reveal consistent performance degradation across both SBP and DBP predictions, regardless of whether

TABLE VI: Ablation results of core modules.

Dataset	Method	MAE (mmHg)		SD (mmHg)		AAMI SP10		BHS Grade	
		SBP	DBP	SBP	DBP	SBP	DBP	SBP	DBP
Nature	TransBP	4.75	4.26	6.08	5.53	✓	✓	A	A
	variant T1	6.67	4.96	8.45	6.16	✓	✓	C	B
	variant T2	5.70	4.46	7.28	5.68	✓	✓	B	A
Self-built	TransBP	5.02	4.43	6.05	5.61	✓	✓	B	A
	variant T1	6.30	5.34	7.61	6.51	✓	✓	C	B
	variant T2	5.63	4.79	7.06	5.78	✓	✓	B	B

TABLE VII: Impact of different fine-tuning strategies.

Dataset	Fine-tuning Strategy	MAE (mmHg)		SD (mmHg)		AAMI SP10		BHS Grade	
		SBP	DBP	SBP	DBP	SBP	DBP	SBP	DBP
Nature	Full Strategy	4.75	4.26	6.08	5.53	✓	✓	A	A
	Stage 1	5.92	4.65	7.48	5.75	✓	✓	C	A
	Stage 1 + Stage 2	9.14	4.59	7.39	5.91	✓	✓	D	A
	Stage 1 + Stage 3	6.32	4.71	7.50	5.98	✓	✓	C	A
Self-built	Full Strategy	5.02	4.43	6.05	5.61	✓	✓	B	A
	Stage 1	5.82	4.68	7.34	5.87	✓	✓	B	A
	Stage 1 + Stage 2	8.36	4.83	7.53	5.69	✓	✓	D	B
	Stage 1 + Stage 3	6.14	4.70	7.46	5.92	✓	✓	C	B

on the Nature dataset or the self-built dataset. These results indicate that the cross-modal attention module effectively captures the complementary relationship between primary and auxiliary signals, facilitating feature fusion and improving the accuracy of BP prediction.

D. Fine-tuning Strategy Analysis (RQ3)

1) *Impact of Different Fine-tuning Strategies:* To verify the rationality and necessity of the proposed three-stage fine-tuning strategy, we conduct experiments to analyze the impact of different combinations of stages on BP prediction performance. Specifically, we compare the following four fine-tuning configurations:

- Full strategy: The complete three-stage fine-tuning pipeline proposed in this paper.
- Stage 1 only: Joint fine-tuning of the feature encoder and two regressors with MMD-based marginal distribution alignment.
- Stage 1 + Stage 2: Stage 1 followed by the maximization of prediction discrepancy between the two regressors on unlabeled target data.
- Stage 1 + Stage 3: Stage 1 followed by adapting the feature extractor to minimize the discrepancy between the regressors without going through the discrepancy maximization stage.

The experimental results are shown in Table VII. Our observations from Table VII are as follows.

First, the TransBP model, when only performing fine-tuning of stage 1, has limited performance. Stage 1 involves supervising with labeled source and target samples and performing marginal distribution alignment using MMD to enable basic cross-domain modeling. However, the role of stage 1 is heavily restricted due to the limited number of labeled samples in the target domain.

Second, TransBP that is fine-tuned using the strategy of “Stage 1 + Stage 2” exhibits the worst performance. Compared

to TransBP using the full fine-tuning strategy, its MAE in SBP increases by 92.4% on the Nature dataset and by 66.5% on the self-built dataset. Meanwhile, TransBP using the strategy of “Stage 1 + Stage 3” also shows a severe degradation in performance. These results indicate that Stage 2 and Stage 3, each of which is for unlabeled data in the target domain to capture the feature representation of samples of the target domain, should be employed jointly to improve performance.

Lastly, TransBP using the full fine-tuning strategy yields the best performance across different datasets compared to TransBP using other fine-tuning configurations. This confirms the necessity and effectiveness of the three-stage fine-tuning strategy.

2) *Impact of the Number of Labeled Training Samples in the Target Domain:* Collecting large-scale synchronized radar signals with labeled BP values is time-consuming and costly. In contrast, merely acquiring unlabeled radar signals is a low-cost and feasible process. TransBP capitalizes on this distinction: it leverages datasets with abundant labeled physiological signals during the pre-training phase to capture fundamental BP-related features, then fine-tunes the model using a small number of labeled radar-based samples as well as unlabeled radar signals during different stages of the fine-tuning phase. This design allows TransBP to fully utilize the available unlabeled radar signals as auxiliary information while efficiently learning from the scarce labeled synchronized data, thereby achieving competitive performance even with limited labeled samples in the target domain. This highlights its capability to address the data scarcity challenge in contactless BP monitoring.

To evaluate the impact of the number of labeled training samples in the target domain on BP prediction performance, we conduct experiments using three different proportions: 5%, 10%, and 20%. The results are summarized in Table VIII. As expected, the BP prediction performance of TransBP improves with an increasing proportion of labeled samples. But criti-

TABLE VIII: Effect of ratio of labeled samples.

Labeled Sample Ratio	MAE (mmHg)		SD (mmHg)		BHS	
	SBP	DBP	SBP	DBP	SBP	DBP
Nature Dataset						
5%	5.02	4.56	6.39	5.47	B	A
10%	4.96	4.54	6.38	5.44	B	A
20%	4.75	4.26	6.08	5.53	A	A
Self-built Dataset						
5%	5.20	4.60	6.25	6.77	B	A
10%	5.11	4.47	6.18	5.69	B	A
20%	5.02	4.43	6.05	5.61	B	A

TABLE IX: Effect of kernel size.

Dataset	K	MAE (mmHg)		SD (mmHg)	
		SBP	DBP	SBP	DBP
Nature	3	5.23	5.54	6.14	6.35
	5	5.12	5.26	6.04	6.31
	7	4.91	5.02	6.07	6.18
	9	4.94	4.42	6.09	5.84
	11	4.83	4.21	6.10	5.61
	13	4.75	4.26	6.08	5.53
	15	4.77	4.29	6.11	5.70
	17	4.92	4.48	6.31	5.86
Self-built	3	5.36	4.97	6.85	5.93
	5	5.30	4.91	6.81	5.87
	7	5.39	4.76	6.67	5.72
	9	5.07	4.59	6.43	5.68
	11	5.09	4.50	6.21	5.71
	13	5.02	4.43	6.05	5.61
	15	5.12	4.41	6.07	5.65
	17	5.37	4.76	6.43	5.78

cally, even when using only 5% or 10% of the labeled target-domain samples, it maintains performance levels close to those achieved with 20% labeled samples. Moreover, the BHS grades remain relatively stable across all three proportions, confirming that TransBP can deliver reliable results even when labeled synchronized samples are limited. This underscores its practical value for scenarios where large-scale radar signal-BP pairs are difficult to obtain.

E. Hyperparameter Sensitivity Analysis (RQ4)

1) Effect of Kernel Size on BP Prediction Performance:

To evaluate the effect of kernel size (K) in the multimodal embedding module on performance, we conduct experiments by setting different kernel sizes and calculating the MAE and SD of BP prediction values. As shown in Table IX, the results indicate that the BP prediction performance generally improves when utilizing larger convolution kernel sizes.

Given that the sampling rate of the dataset collected in this study is 125 Hz, a complete pulse cycle typically spans approximately 75–125 sampling points. By employing a larger kernel size (e.g., 11, 13, or 15) in each convolution layer, the multimodal embedding module can encompass an entire pulse waveform within its receptive field. This enables the model to effectively capture the temporal structure of the waveform and its relationship with BP, thereby improving prediction accuracy.

TABLE X: Effect of number of attention heads.

Dataset	h	MAE (mmHg)		SD (mmHg)	
		SBP	DBP	SBP	DBP
Nature	4	5.24	6.75	6.38	7.71
	8	4.75	4.26	6.08	5.53
	16	4.71	5.27	6.14	5.89
Self-built	4	5.91	4.67	6.64	5.73
	8	5.02	4.43	6.05	5.61
	16	5.27	5.02	6.43	5.75

TABLE XI: Effect of number of blocks.

Dataset	L	MAE (mmHg)		SD (mmHg)	
		SBP	DBP	SBP	DBP
Nature	1	6.07	5.69	7.97	7.34
	2	5.51	5.38	7.28	6.95
	4	4.75	4.26	6.08	5.53
	8	9.20	5.45	6.36	7.01
Self-built	1	5.54	5.20	6.75	5.74
	2	5.03	4.65	6.36	5.67
	4	5.02	4.43	6.05	5.61
	8	5.23	4.61	6.54	5.57

2) *Effect of Number of Attention Heads on BP Prediction Performance:* To evaluate the impact of the number of attention heads (h) in the multi-head attention mechanism on BP prediction performance, we conduct experiments using three configurations: 4, 8, and 16 heads. As shown in Table X, the results demonstrate that setting the number of attention heads to 8 yields the best overall performance on both datasets in terms of MAE and SD. This configuration significantly outperforms the other two settings.

3) *Effect of Number of Blocks on BP Prediction Performance:* To evaluate the influence of the number of blocks (L) in the cross-modal attention module on BP prediction performance, we evaluate the model under four configurations: 1, 2, 4, and 8 blocks. As shown in Table XI, the model with 4 blocks achieves the best performance across both datasets in terms of MAE and SD, and consistently outperforms the other settings.

4) *Effect of Loss Weight Ratio of SBP to DBP:* To evaluate the impact of the loss weight ratio between SBP and DBP (λ) on model performance, five different weight ratios are tested. As shown in Table XII, increasing the SBP weight gradually improves SBP prediction accuracy, while DBP slightly decreases. These results indicate that the choice of loss weight ratio does not yield a clearly superior setting, but rather reflects a trade-off between SBP and DBP prediction. While the performance varies across different ratios, the differences are relatively moderate, indicating that the model is not strongly dependent on a specific ratio.

F. Robustness Analysis (RQ5)

1) *Impact of Subject Diversity:* We conduct a subgroup analysis to assess the impact of subject diversity. Specifically, subjects are divided into three age categories: young (< 44), middle-aged (44–59), and elderly (≥ 60), and three BMI categories: light (< 18.5), medium (18.5–23.9), and heavy ($\geq$

TABLE XII: Effect of loss weight ratio of SBP to DBP.

Dataset	λ	MAE (mmHg)		SD (mmHg)	
		SBP	DBP	SBP	DBP
Nature	1:1	4.91	4.18	6.37	5.49
	3:1	4.85	4.22	6.23	5.52
	6:1	4.82	4.17	6.11	5.57
	9:1	4.75	4.26	6.08	5.53
	12:1	4.69	4.48	6.05	5.71
Self-built	1:1	5.21	4.37	6.32	5.59
	3:1	5.13	4.44	6.17	5.58
	6:1	5.05	4.42	6.09	5.63
	9:1	5.02	4.43	6.05	5.61
	12:1	4.96	4.66	6.01	5.84

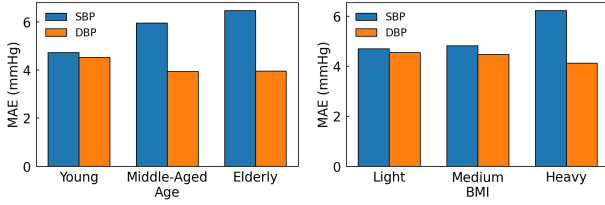


Fig. 9: Impact of age.

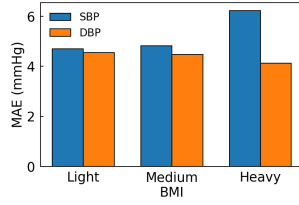


Fig. 10: Impact of BMI.

24). Figure 9 shows the MAEs of SBP and DBP predictions under different ages on the self-built dataset. It can be seen that the SBP prediction error increases with age. This may be because older subjects tend to have higher SBP, making it harder to predict accurately. Figure 10 shows the MAEs of SBP and DBP predictions under different BMIs on the self-built dataset. As the subject's BMI increases, the BP prediction error slowly increases. This may be due to the fact that the received signals from subjects with higher BMI tend to have lower signal-to-noise ratios.

2) *Impact of Subject-device Distance*: To evaluate the system performance at different subject-device distances, we vary the distance between the UWB device and the subject from 0.3 m to 1.5 m at a step size of 0.3 m. As shown in Figure 11, the MAEs of SBP predictions are 3.31 mmHg, 5.17 mmHg, 4.32 mmHg, 5.02 mmHg, and 7.18 mmHg, respectively, while the MAEs of DBP predictions are 1.41 mmHg, 1.98 mmHg, 2.36 mmHg, 4.43 mmHg, and 5.37 mmHg, respectively. The results show a clear degradation in accuracy as the distance increases. This is because increasing the distance can reduce the received signal strength and SNR, which in turn increases BP estimated errors.

V. LIMITATIONS AND DISCUSSION

In this work, we propose a transfer learning-enabled deep learning model to achieve BP prediction. We briefly discuss the limitations and some promising future research directions below.

Body motion. TransBP is evaluated under static conditions, since body motion can introduce non-stationary disturbances (e.g., baseline drift, signal fluctuations, and waveform distortion). Although TransBP requires only a stationary period of 5 seconds for BP prediction, it still affects the integrity of continuous BP monitoring. Future work can model and mitigate motion effects to enable motion-tolerant BP prediction.

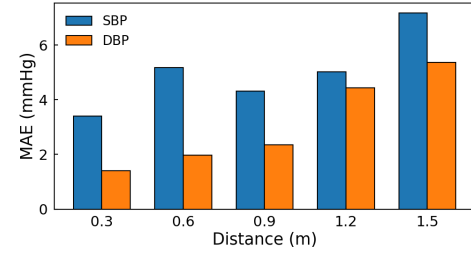


Fig. 11: Impact of Subject-device Distance.

Data distribution. The BP distributions in both the Nature dataset and the self-built dataset exhibit a long-tail effect. To assess the impact of data imbalance, we report the performance across different BP ranges in Table IV, showing lower accuracy for subjects with abnormal BP than for those with normal BP. Nevertheless, the prediction accuracy remains within an acceptable range. Collecting more hypertensive and hypotensive samples is expected to further improve prediction accuracy.

VI. CONCLUSION

In this paper, we collect wireless signals reflected from the human body and predicts BP values. The most distinctive feature of our developed system lies in the fact that we design and train a transfer learning-enabled deep learning model TransBP. TransBP is pretrained on data from the source domain, i.e., physiological signals from wearable devices, and then fine-tuned on data from the target domain, i.e., limited labeled radar samples and a large quantity of unlabeled radar samples through a carefully-designed three-stage process. The good performance results of TransBP show that our work offers a promising technical framework for contactless BP prediction.

Furthermore, with the ongoing miniaturization and cost reduction of radar devices, new radar devices with different antenna configurations and signal properties are constantly appearing. This diversity requires adopting tailored signal processing procedures for reflected radar signals, creating substantial barriers in the development of contactless sensing systems. Our work demonstrates a practical and viable framework for achieving high prediction accuracy: it simply needs to collect a few labeled radar samples, while confining the discrepancies among radar devices to the process of extracting physiological signal waveforms from radar signals. Such a framework enables the application of new radar devices to contactless BP prediction at a low cost, and even to wireless monitoring of other vital signs.

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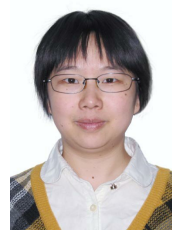
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